

HILLOL DAS PIAS

AI-Native Full-Stack Engineer • 4.5+ Years Experience

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[Portfolio](#) • [GitHub](#) • [LinkedIn](#)

PROFESSIONAL SUMMARY

AI-native full-stack engineer with 4.5+ years building production web applications and a growing specialisation in LLM-powered product development. Hands-on with the Anthropic and OpenAI APIs, LangChain, LangGraph, RAG pipelines (ChromaDB), tool-calling AI agents, and multi-agent systems — including a deployed ReAct agent orchestrating 17 tools with LangSmith tracing. Proficient across the modern JavaScript ecosystem (Next.js, Node.js, Express.js, React, TypeScript) and Python backends (FastAPI, async SQLAlchemy). Proven delivery across RESTful APIs, real-time systems (Socket.IO), headless CMS architectures, e-commerce platforms, and containerised services via Docker and CI/CD pipelines. Collaborative and self-driven, with experience across the full development lifecycle in remote-first environments.

CORE COMPETENCIES

AI / LLM LangChain • LangGraph • RAG pipelines (ChromaDB) • Multi-agent & ReAct agents • Anthropic & OpenAI APIs • Tool use & function calling • LLM streaming (SSE) • LangSmith tracing • Prompt engineering

Full-Stack Next.js • React • Node.js/Express • TypeScript • FastAPI • Socket.IO • REST APIs • Java (Spring Boot)

Infra & Data Docker • GitHub Actions CI/CD • PostgreSQL • MySQL • Neon • async SQLAlchemy 2.0 • Alembic • MongoDB • ChromaDB (vector DB) • Vercel • Render • Railway • Stripe • Strapi CMS • Supabase

System Design & Deployment Redis (caching) • Load Balancing • CDN • Message Queues • Read Replicas • Object Storage • AWS

PROFESSIONAL EXPERIENCE

Full-Stack Developer | SwaPE IT Solutions

Dec 2023 – Present

- Architected and shipped production web apps using Next.js (App Router), Node.js, React, and Python (FastAPI) services, with fully responsive UIs via Tailwind CSS and Framer Motion.
- Led system design decisions and defined scalable architecture patterns across multiple client projects.
- Integrated LLM APIs (Anthropic, OpenAI) into client-facing features, gaining hands-on experience with prompt engineering and AI-powered product flows.
- Improved code quality through automated tooling, linting, peer reviews, and performance profiling.
- Integrated third-party services and APIs, reducing development time by establishing reusable integration patterns.

Full-Stack Developer | Raj IT Solutions Ltd.

Nov 2021 – Dec 2023

- Built RESTful APIs with Next.js, Node.js (Express), and Python (FastAPI); implemented real-time features using Socket.IO serving concurrent users without performance degradation.
- Developed and maintained backend microservices (including Java/Spring Boot) ensuring high availability and horizontal scalability.
- Integrated Strapi headless CMS, enabling non-technical stakeholders to manage content without developer intervention.
- Established CI/CD pipelines with Docker, cutting deployment time significantly and reducing production incidents.

Freelance Full-Stack Developer | Independent

Oct 2015 – Present

- Building and shipping AI-agent and AI-assisted products (see AI Projects below), spanning Next.js frontends, FastAPI/Node backends, and LangChain/LangGraph agent orchestration.
- Delivered e-commerce and booking systems using Node.js, React, and Next.js with Stripe payment integration, role-based access control, and RESTful APIs.
- Designed and optimised PostgreSQL, MongoDB, and MySQL databases for scalable, production-ready applications.

Graphic Designer | Cloudville

Jan 2020 – Oct 2021

- Designed digital assets and social media content; developed strong visual communication and client-facing skills directly applicable to frontend UI work.

AI PROJECTS

Ledger — AI Agent Task & Project Manager • LangGraph • FastAPI • async SQLAlchemy • PostgreSQL • ChromaDB • LangSmith

- Built a LangGraph-orchestrated ReAct agent that reasons over and selects from 17 tools in a loop — creating/updating tasks and projects and answering questions grounded in live database retrieval (RAG via ChromaDB) rather than static prompting.
- Streamed agent responses token-by-token to the frontend via `astream_events`, with every run traced end-to-end in LangSmith for observability and debugging.
- Backend built on FastAPI, async SQLAlchemy 2.0, and PostgreSQL (Neon) with Alembic migrations; deployed on Render alongside a separate Next.js frontend.

Live demo: taskapi-fronend.vercel.app • Backend: github.com/HillolPias/taskapi-backend

Cookmind — AI Recipe Generator • Next.js • Google Gemini API • Strapi • PostgreSQL

- Built an AI recipe generator that analyses a photo of a user's ingredients via the Google Generative AI (Gemini) API and returns tailored, ready-to-cook recipes, with drag-and-drop image upload and PDF export of results.
- Implemented Clerk-based authentication and Arcjet bot/rate-limit protection on a Next.js 16 (App Router) + React 19 frontend, backed by a Strapi 5 headless CMS on PostgreSQL.

Live demo: cookmind-ivory.vercel.app • GitHub: github.com/HillolPias/cookmind

ADDITIONAL PROJECTS

Chatty App — real-time full-stack chat application (JavaScript). [GitHub](#)

Product Store (MERN) — full-stack e-commerce product catalog. [GitHub](#)

NextLevel Food/Foodies — recipe sharing application. [GitHub](#)

EDUCATION

BSc, Applied Physics & Electronic Engineering — University of Rajshahi

COURSES & CERTIFICATIONS

Agentic AI Engineering with LangChain & LangGraph (Udemy) • Functions, Tools & Agents with LangChain (DeepLearning.AI) • React — incl. Next.js, Redux (Udemy) • FastAPI Complete Course (Udemy) • Full-Stack Web Dev Bootcamp (Udemy) • CCNA — RUET